

Lesson 6-1: Division Expressions with Fractions and Whole Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $8.4 \div 4$.

Answer: 2.1

84 tenths divided by 4 is 21 tenths.

Prerequisite skill: Divide decimals and whole numbers

TASK 2 • FOUNDATION 2

What is the reciprocal of 5?

Answer: $1/5$

Write 5 as $5/1$, then exchange numerator and denominator.

Prerequisite skill: Name the reciprocal of a whole number and of a unit fraction

TASK 3 • FOUNDATION 3

Compute $12 \times 2/3$.

Answer: 8

Divide 12 by 3, then multiply by 2.

Prerequisite skill: Multiply a whole number by a fraction

TASK 4 • FOUNDATION 4

Write $15/3$ as a division expression and evaluate it.

Answer: $15 \div 3 = 5$

A fraction bar means divide the numerator by the denominator.

Prerequisite skill: Read a fraction bar as "divided by"

Lesson 6-1: Division Expressions with Fractions and Whole Numbers

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Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$4.8 \div 2$? How many fourths are in 3? Reciprocal of 5?

Answer: $4.8 \div 2 = 2.4$. There are 12 fourths in 3 (since $3 \div 1/4 = 3 \times 4 = 12$). Reciprocal of 5 is $1/5$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$6.4 \div 2$? How many thirds are in 4? Reciprocal of 8?

Answer: $6.4 \div 2 = 3.2$. There are 12 thirds in 4 ($4 \div 1/3 = 4 \times 3 = 12$). Reciprocal of 8 is $1/8$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$9.6 \div 4$? How many halves are in 5? Reciprocal of $1/6$?

Answer: $9.6 \div 4 = 2.4$. There are 10 halves in 5. Reciprocal of $1/6$ is 6.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A ribbon is 6 feet long. How many $3/4$ -foot pieces can be cut from it?

Answer: $6 \div 3/4 = 6 \times 4/3 = 24/3 = 8$ pieces.

Check the result against the situation and explain your reasoning.

Lesson 6-2: Division Expressions with Fractions and Mixed Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Write $2\frac{3}{4}$ as an improper fraction.

Answer: $\frac{11}{4}$

Two wholes contain 8 fourths; add 3 fourths to get 11 fourths.

Prerequisite skill: Rewrite a mixed number as an improper fraction

TASK 2 • FOUNDATION 2

Compute $\frac{2}{3} \times \frac{3}{5}$.

Answer: $\frac{2}{5}$

Multiply to get $\frac{6}{15}$, then simplify by dividing both parts by 3.

Prerequisite skill: Multiply two fractions

TASK 3 • FOUNDATION 3

Find the reciprocal of $\frac{4}{7}$.

Answer: $\frac{7}{4}$

Exchange numerator and denominator; their product will be 1.

Prerequisite skill: Find the reciprocal of any fraction

TASK 4 • FOUNDATION 4

Simplify $\frac{18}{24}$.

Answer: $\frac{3}{4}$

Divide numerator and denominator by their greatest common factor, 6.

Prerequisite skill: Simplify a fraction to lowest terms

Lesson 6-2: Division Expressions with Fractions and Mixed Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

2 $\frac{3}{4}$ as an improper fraction? Reciprocal of $\frac{3}{5}$?
 $\frac{2}{3} \times \frac{5}{4}$?

Answer: 2 $\frac{3}{4}$ as improper fraction: $(2 \times 4 + 3) / 4 = 11/4$. Reciprocal of $\frac{3}{5}$ is $\frac{5}{3}$. Product: $\frac{2}{3} \times \frac{5}{4} = 10/12 = 5/6$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

3 $\frac{2}{5}$ as an improper fraction? Reciprocal of $\frac{4}{7}$?
 $\frac{3}{4} \times \frac{2}{5}$?

Answer: 3 $\frac{2}{5} = (3 \times 5 + 2)/5 = 17/5$. Reciprocal of $\frac{4}{7}$ is $\frac{7}{4}$. $\frac{3}{4} \times \frac{2}{5} = 6/20 = 3/10$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

1 $\frac{5}{6}$ as an improper fraction? Reciprocal of $\frac{9}{2}$?
 $\frac{5}{6} \times \frac{3}{10}$?

Answer: 1 $\frac{5}{6} = 11/6$. Reciprocal of $\frac{9}{2}$ is $\frac{2}{9}$. $\frac{5}{6} \times \frac{3}{10} = 15/60 = 1/4$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

How many 1 $\frac{1}{2}$ -cup servings are in 6 cups?

Answer: $6 \div \frac{3}{2} = 6 \times \frac{2}{3} = 4$ servings.

Check the result against the situation and explain your reasoning.

Lesson 6-3: Explore Numerical Expressions with Exponents

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $3 \times 3 \times 3 \times 3$.

Answer: 81

Successive products are 9, 27, and 81.

Prerequisite skill: Multiply the same factor repeatedly

TASK 2 • FOUNDATION 2

Evaluate 5^2 .

Answer: 25

5^2 means 5×5 .

Prerequisite skill: Recall 2^2 , 3^2 , 4^2 , 5^2 , 2^3 , and 3^3

TASK 3 • FOUNDATION 3

Evaluate $10 \times 10 \times 10$.

Answer: 1000

Three factors of 10 give one thousand.

Prerequisite skill: Multiply by 10 repeatedly to build powers of 10

TASK 4 • FOUNDATION 4

How many factors of 2 appear in $2 \times 2 \times 2 \times 2 \times 2$?

Answer: 5

Count each repeated factor, not the multiplication signs.

Prerequisite skill: Count how many factors are written

Lesson 6-3: Explore Numerical Expressions with Exponents

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

6^2 ? Write $2 \times 2 \times 2 \times 2$ as a power. 10^3 ?

Answer: $6^2 = 6 \times 6 = 36$. $2 \times 2 \times 2 \times 2$ as a power = 2^4 . $10^3 = 10 \times 10 \times 10 = 1,000$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

7^2 ? Write $3 \times 3 \times 3 \times 3 \times 3$ as a power. 10^4 ?

Answer: $7^2 = 49$. $3 \times 3 \times 3 \times 3 \times 3 = 3^5$. $10^4 = 10,000$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

4^3 ? Write $5 \times 5 \times 5$ as a power. 10^2 ?

Answer: $4^3 = 64$. $5 \times 5 \times 5 = 5^3$. $10^2 = 100$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Which is greater, 3^4 or 4^3 ?

Answer: $3^4 = 81$ and $4^3 = 64$, so 3^4 is greater.

Check the result against the situation and explain your reasoning.

Lesson 6-4: Write and Evaluate Numerical Expressions with Exponents

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.2.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Evaluate 4^3 .

Answer: 64

$4 \times 4 \times 4 = 16 \times 4 = 64$.

Prerequisite skill: Evaluate a power

TASK 2 • FOUNDATION 2

Evaluate $2 + 3^2 \times 4$.

Answer: 38

Exponent first gives 9; multiplication gives 36; then add 2.

Prerequisite skill: Follow order of operations with parentheses and exponents

TASK 3 • FOUNDATION 3

Evaluate $(8 + 4)/3$.

Answer: 4

The fraction bar groups the numerator: $12 \div 3 = 4$.

Prerequisite skill: Read a fraction bar as a grouping symbol

TASK 4 • FOUNDATION 4

Write an expression for 7 less than the product of 3 and 5.

Answer: $3 \times 5 - 7$

Start with the product, then subtract 7. Its value is 8.

Prerequisite skill: Translate "sum," "product," and "less than" into symbols

Lesson 6-4: Write and Evaluate Numerical Expressions with Exponents

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.2.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$4 + 6 \div 2$? $3 + 2^2 \times 5$? Which operation comes first?

Answer: $4 + 6 \div 2$: Divide first ($6 \div 2 = 3$), then add: $4 + 3 = 7$. $3 + 2^2 \times 5$: Exponent first ($2^2 = 4$), then multiply ($4 \times 5 = 20$), then add: $3 + 20 = 23$. Exponent comes first, then multiplication, then addition.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$8 + 12 \div 4$? $5 + 3^2 \times 2$? Which operation comes first?

Answer: $8 + 12 \div 4$: divide first (3), then add: 11. $5 + 3^2 \times 2$: exponent (9), then multiply (18), then add: 23. The exponent comes first.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$20 - 6 \times 2$? $4 + 2^3 \div 4$? Which operation comes first?

Answer: $20 - 6 \times 2$: multiply first (12), then subtract: 8. $4 + 2^3 \div 4$: exponent (8), then divide (2), then add: 6. The exponent comes first.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Evaluate $2 \times (3 + 4)^2 - 10$.

Answer: Parentheses: 7. Exponent: 49. Multiply: 98. Subtract: 88.

Check the result against the situation and explain your reasoning.

Lesson 6-5: Write and Evaluate Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.2

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Evaluate $n + 8$ when $n = 6$.

Answer: 14

Replace n with 6: $6 + 8 = 14$.

Prerequisite skill: Substitute a number for a letter, then compute

TASK 2 • FOUNDATION 2

Evaluate $2x + 3$ when $x = 5$.

Answer: 13

Substitute, then multiply before adding: $2 \times 5 + 3$.

Prerequisite skill: Follow order of operations after substituting

TASK 3 • FOUNDATION 3

Evaluate $4n$ when $n = 7$.

Answer: 28

A number next to a variable means multiplication: 4×7 .

Prerequisite skill: Read $4n$ as $4 \times n$

TASK 4 • FOUNDATION 4

Write an expression for 9 more than a number n .

Answer: $n + 9$

More than indicates adding 9 to the starting number.

Prerequisite skill: Translate a word phrase into an expression

Lesson 6-5: Write and Evaluate Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.2

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Evaluate $4n + 3$ when $n = 5$. Write an expression for "7 more than a number."

Answer: When $n = 5$: $4(5) + 3 = 20 + 3 = 23$. Expression for '7 more than a number': $n + 7$ (or $x + 7$).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Evaluate $3n + 8$ when $n = 6$. Write an expression for "5 less than a number."

Answer: $3(6) + 8 = 18 + 8 = 26$. Expression: $n - 5$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Evaluate $2n - 7$ when $n = 10$. Write an expression for "the product of 6 and a number."

Answer: $2(10) - 7 = 20 - 7 = 13$. Expression: $6n$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Write an expression for the cost of n tickets at \$12 each plus a \$5 fee, then find the cost for 4 tickets.

Answer: $12n + 5$. At $n = 4$: $12(4) + 5 = 53$ dollars.

Check the result against the situation and explain your reasoning.

Lesson 6-6: Identify Equivalent Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Simplify $5x + 2x$.

Answer: $7x$

Both terms count x units; combine coefficients $5 + 2$.

Prerequisite skill: Combine like terms such as $5x + 2x$

TASK 2 • FOUNDATION 2

Evaluate $3x + 4$ when $x = 2$.

Answer: 10

Substitute 2, then compute $6 + 4$.

Prerequisite skill: Evaluate an expression at a chosen value

TASK 3 • FOUNDATION 3

Evaluate $3(4 + 5)$ using distribution.

Answer: 27

$3 \times 4 + 3 \times 5 = 12 + 15$.

Prerequisite skill: Apply the distributive property with numbers, e.g. $3(4 + 5)$

TASK 4 • FOUNDATION 4

Compare $2(x + 3)$ and $2x + 6$ at $x = 4$. Give both values.

Answer: 14 and 14

$2(4 + 3) = 14$ and $2 \times 4 + 6 = 14$. Matching at one value alone does not prove equivalence for all values.

Prerequisite skill: Compare two results to decide whether expressions match

Lesson 6-6: Identify Equivalent Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$5x + 2x = ?$ Evaluate $2(x + 3)$ and $2x + 6$ at $x = 4$
— are they the same?

Answer: $5x + 2x = 7x$. At $x = 4$: $2(4 + 3) = 2(7) = 14$; and $2(4) + 6 = 8 + 6 = 14$. Yes, both equal 14 at $x = 4$. The distributive property, $2(x + 3) = 2x + 6$, shows they are equivalent for every x .

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$8y + 3y = ?$ Evaluate $3(x + 2)$ and $3x + 6$ at $x = 5$
— are they the same?

Answer: $8y + 3y = 11y$. At $x = 5$: $3(7) = 21$ and $15 + 6 = 21$. Yes — the distributive property shows they match for every x .

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$9a - 4a = ?$ Evaluate $4(x + 1)$ and $4x + 4$ at $x = 3$
— are they the same?

Answer: $9a - 4a = 5a$. At $x = 3$: $4(4) = 16$ and $12 + 4 = 16$. Yes — equivalent for every x .

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Are $3(2x + 4)$ and $6x + 7$ equivalent? Justify.

Answer: No. Distributing gives $6x + 12$, not $6x + 7$. At $x = 0$ they give 12 and 7 — different.

Check the result against the situation and explain your reasoning.

Lesson 6-7: Find Factors and Multiples

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $84 \div 7$.

Answer: 12

The related fact is $7 \times 12 = 84$.

Prerequisite skill: Recall multiplication and division facts

TASK 2 • FOUNDATION 2

List every positive factor of 18.

Answer: 1, 2, 3, 6, 9, 18

Use factor pairs 1×18 , 2×9 , and 3×6 .

Prerequisite skill: List every factor pair of a number up to 100

TASK 3 • FOUNDATION 3

List the first four positive multiples of 7.

Answer: 7, 14, 21, 28

Multiply 7 by 1, 2, 3, and 4.

Prerequisite skill: List the first several multiples of a number

TASK 4 • FOUNDATION 4

Which of 2, 3, 5, 9, and 10 divide 135 evenly?

Answer: 3, 5, 9

Digit sum 9 makes it divisible by 3 and 9; final digit 5 makes it divisible by 5. It is not even and does not end in 0.

Prerequisite skill: Test divisibility by 2, 3, 5, 9, and 10

Lesson 6-7: Find Factors and Multiples

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

All the factors of 18? First five multiples of 6?
GCF of 18 and 24?

Answer: Factors of 18: 1, 2, 3, 6, 9, 18. First five multiples of 6: 6, 12, 18, 24, 30. GCF of 18 and 24: Common factors are 1, 2, 3, 6; Greatest is 6.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

All the factors of 24? First five multiples of 8?
GCF of 24 and 36?

Answer: Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Multiples of 8: 8, 16, 24, 32, 40. $\text{GCF}(24, 36) = 12$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

All the factors of 20? First five multiples of 5?
GCF of 20 and 30?

Answer: Factors of 20: 1, 2, 4, 5, 10, 20. Multiples of 5: 5, 10, 15, 20, 25. $\text{GCF}(20, 30) = 10$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

You have 24 pencils and 36 erasers to split into identical kits with nothing left over. What is the largest number of kits?

Answer: $\text{GCF}(24, 36) = 12$ kits, each with 2 pencils and 3 erasers.

Check the result against the situation and explain your reasoning.

Lesson 6-8: Generate Equivalent Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Find the greatest common factor of 18 and 30.

Answer: 6

Their common factors are 1, 2, 3, and 6; choose the greatest.

Prerequisite skill: Find the GCF of two numbers

TASK 2 • FOUNDATION 2

Factor $12x + 20$ using the greatest common factor.

Answer: $4(3x + 5)$

Both terms share 4; divide each term by 4 inside the parentheses.

Prerequisite skill: Apply the distributive property forward and backward

TASK 3 • FOUNDATION 3

Simplify $4a + 3a + 2$.

Answer: $7a + 2$

Combine the a terms; the constant 2 is a different kind of term.

Prerequisite skill: Combine like terms

TASK 4 • FOUNDATION 4

Compute $6(10 + 4)$.

Answer: 84

Distribute: $60 + 24 = 84$.

Prerequisite skill: Multiply a number by a sum

Lesson 6-8: Generate Equivalent Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

GCF of 18 and 24? Expand $3(x + 5)$. Factor $4x + 12$.

Answer: $\text{GCF}(18, 24) = 6$. Expand: $3(x + 5) = 3x + 15$. Factor: $4x + 12 = 4(x + 3)$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

GCF of 20 and 30? Expand $5(x + 4)$. Factor $6x + 18$.

Answer: $\text{GCF}(20, 30) = 10$. $5(x + 4) = 5x + 20$. $6x + 18 = 6(x + 3)$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

GCF of 12 and 16? Expand $4(x + 7)$. Factor $9x + 15$.

Answer: $\text{GCF}(12, 16) = 4$. $4(x + 7) = 4x + 28$. $9x + 15 = 3(3x + 5)$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Factor $24x + 36$ completely, then check by expanding.

Answer: $\text{GCF}(24, 36) = 12$, so $24x + 36 = 12(2x + 3)$. Check: $12(2x) + 12(3) = 24x + 36$.
✓

Check the result against the situation and explain your reasoning.

Lesson 6-9: Divide Whole Numbers by Fractions

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

How many $\frac{1}{5}$ s are in 3?

Answer: 15

Each whole holds 5 fifths, so 3 wholes hold $3 \times 5 = 15$ fifths.

Prerequisite skill: Count how many unit fractions make one whole

TASK 2 • FOUNDATION 2

Compute $9 \times \frac{2}{3}$.

Answer: 6

$9 \times 2 = 18$ thirds, and $18/3 = 6$.

Prerequisite skill: Multiply a whole number by a fraction

TASK 3 • FOUNDATION 3

Write the reciprocal of $\frac{2}{7}$.

Answer: $\frac{7}{2}$

Swap numerator and denominator: $\frac{2}{7} \times \frac{7}{2} = 1$.

Prerequisite skill: Write the reciprocal of a fraction or whole number

TASK 4 • FOUNDATION 4

There are 5 cups of trail mix. Each bag holds $\frac{2}{3}$ cup. How many full bags can be filled, and what is left over? Explain.

Answer: $5 \div \frac{2}{3} = 7 \frac{1}{2}$, so 7 full bags can be filled with half a bag's amount ($\frac{1}{3}$ cup) left over.

Compute $5 \times \frac{3}{2} = 7 \frac{1}{2}$, then decide what the half group means in the story.

Prerequisite skill: Interpret a leftover part of a group in a word problem

Lesson 6-9: Divide Whole Numbers by Fractions

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

How many $\frac{1}{4}$ s are in 1? Reciprocal of $\frac{3}{5}$? $6 \times \frac{3}{4}$?

Answer: There are 4 fourths in 1. The reciprocal of $\frac{3}{5}$ is $\frac{5}{3}$. $6 \times \frac{3}{4} = \frac{18}{4} = 4 \frac{1}{2}$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

How many $\frac{1}{3}$ s are in 1? Reciprocal of $\frac{2}{7}$? $8 \times \frac{3}{4}$?

Answer: There are 3 thirds in 1. The reciprocal of $\frac{2}{7}$ is $\frac{7}{2}$. $8 \times \frac{3}{4} = 6$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

How many $\frac{1}{6}$ s are in 1? Reciprocal of 4? $9 \times \frac{2}{3}$?

Answer: There are 6 sixths in 1. The reciprocal of 4 is $\frac{1}{4}$. $9 \times \frac{2}{3} = 6$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A gardener has 5 bags of soil. Each planter needs $\frac{2}{3}$ of a bag. How many planters can be filled completely, and what part of a bag is left?

Answer: $5 \div \frac{2}{3} = 5 \times \frac{3}{2} = 7 \frac{1}{2}$. Seven planters are filled; $\frac{1}{2}$ of a planter's amount is left, which is $\frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$ of a bag.

Check the result against the situation and explain your reasoning.

Lesson 6-10: Divide Mixed Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Write $4 \frac{2}{3}$ as an improper fraction.

Answer: $14/3$

4 wholes = 12 thirds; add 2 thirds to get $14/3$.

Prerequisite skill: Rewrite a mixed number as an improper fraction

TASK 2 • FOUNDATION 2

Write the reciprocal of $5/8$.

Answer: $8/5$

Swap numerator and denominator: $5/8 \times 8/5 = 1$.

Prerequisite skill: Write the reciprocal of a fraction

TASK 3 • FOUNDATION 3

Multiply $9/4 \times 2/3$.

Answer: $3/2$

$9 \times 2 = 18$ and $4 \times 3 = 12$, so $18/12 = 3/2$.

Prerequisite skill: Multiply two fractions

TASK 4 • FOUNDATION 4

Write $22/4$ as a mixed number.

Answer: $5 \frac{1}{2}$

$22 \div 4 = 5$ with 2 left over: $5 \frac{2}{4} = 5 \frac{1}{2}$.

Prerequisite skill: Rewrite an improper fraction as a whole or mixed number

Lesson 6-10: Divide Mixed Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Write $3\frac{1}{2}$ as an improper fraction. $\frac{7}{2} \times \frac{4}{1}$?
Write $\frac{28}{2}$ as a whole number.

Answer: $3\frac{1}{2} = \frac{7}{2}$ ($3 \times 2 + 1 = 7$). $\frac{7}{2} \times \frac{4}{1} = \frac{28}{2}$. $\frac{28}{2} = 14$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Write $2\frac{1}{4}$ as an improper fraction. $\frac{9}{4} \times \frac{4}{3}$?
Write $\frac{36}{12}$ as a whole number.

Answer: $2\frac{1}{4} = \frac{9}{4}$. $\frac{9}{4} \times \frac{4}{3} = \frac{36}{12}$. $\frac{36}{12} = 3$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Write $4\frac{2}{3}$ as an improper fraction. $\frac{14}{3} \times \frac{3}{2}$?
Write $\frac{42}{6}$ as a whole number.

Answer: $4\frac{2}{3} = \frac{14}{3}$. $\frac{14}{3} \times \frac{3}{2} = \frac{42}{6}$. $\frac{42}{6} = 7$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A painter has $6\frac{2}{3}$ quarts of paint. Each door needs $1\frac{1}{3}$ quarts. How many doors can be painted? Check by multiplying back.

Answer: $6\frac{2}{3} \div 1\frac{1}{3} = \frac{20}{3} \times \frac{3}{4} = \frac{60}{12} = 5$ doors. **Check:** $5 \times 1\frac{1}{3} = 6\frac{2}{3}$ quarts.

Check the result against the situation and explain your reasoning.

Lesson 6-11: Fraction Division Problem Solving

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

A jug holds $\frac{7}{8}$ quart. Each cup holds $\frac{1}{8}$ quart. Write the division equation and name the dividend and the divisor.

Answer: $\frac{7}{8} \div \frac{1}{8} = 7$. The dividend is $\frac{7}{8}$ quart (the total); the divisor is $\frac{1}{8}$ quart (the size of each cup).

The total comes first; the group size is what you divide by.

Prerequisite skill: Name the total (dividend) and the group size (divisor) in a story

TASK 2 • FOUNDATION 2

Compute $\frac{5}{6} \div \frac{1}{3}$.

Answer: $\frac{5}{2}$

$\frac{5}{6} \times \frac{3}{1} = \frac{15}{6} = \frac{5}{2}$, or $2 \frac{1}{2}$.

Prerequisite skill: Divide a fraction by a fraction using the reciprocal

TASK 3 • FOUNDATION 3

Compute $6 \times \frac{1}{8}$.

Answer: $\frac{3}{4}$

$6 \times \frac{1}{8} = \frac{6}{8} = \frac{3}{4}$.

Prerequisite skill: Multiply a whole number by a fraction to check a quotient

TASK 4 • FOUNDATION 4

Without computing, is $\frac{3}{4} \div \frac{1}{4}$ greater than or less than $\frac{3}{4}$? Explain.

Answer: Greater. The divisor $\frac{1}{4}$ is less than 1, so more than one group of $\frac{1}{4}$ fits in $\frac{3}{4}$ (the quotient is 3).

Dividing by a number less than 1 gives a quotient greater than the dividend.

Prerequisite skill: Compare a divisor to 1 to predict whether a quotient is greater or less than the dividend

Lesson 6-11: Fraction Division Problem Solving

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

In 'How many $\frac{1}{8}$ -pound casts can be made from $\frac{7}{8}$ pound?', which number is the dividend? $\frac{7}{8} \div \frac{1}{8}$? $7 \times \frac{1}{8}$?

Answer: The total, $\frac{7}{8}$ pound, is the dividend; $\frac{1}{8}$ is the divisor. $\frac{7}{8} \div \frac{1}{8} = \frac{7}{8} \times \frac{8}{1} = 7$. $7 \times \frac{1}{8} = \frac{7}{8}$, which checks the quotient.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

In 'How many $\frac{1}{6}$ -mile checkpoints fit on a $\frac{5}{6}$ -mile trail?', which number is the dividend? $\frac{5}{6} \div \frac{1}{6}$? $5 \times \frac{1}{6}$?

Answer: $\frac{5}{6}$ mile is the dividend. $\frac{5}{6} \div \frac{1}{6} = 5$. $5 \times \frac{1}{6} = \frac{5}{6}$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

In 'How many $\frac{3}{10}$ -liter vials can be filled from $\frac{9}{10}$ liter?', which number is the dividend? $\frac{9}{10} \div \frac{3}{10}$? $3 \times \frac{3}{10}$?

Answer: $\frac{9}{10}$ liter is the dividend. $\frac{9}{10} \div \frac{3}{10} = 3$. $3 \times \frac{3}{10} = \frac{9}{10}$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A $\frac{3}{4}$ -pound bag of clay is shared equally by 3 students. A second problem asks how many $\frac{3}{4}$ -pound bags make 3 pounds. Write and solve both equations and explain why the answers differ.

Answer: Sharing: $\frac{3}{4} \div 3 = \frac{1}{4}$ pound per student. Grouping: $3 \div \frac{3}{4} = 4$ bags. The total and the divisor trade places, so one quotient is a group size and the other is a number of groups.

Check the result against the situation and explain your reasoning.

Lesson 6-12: Least Common Multiple

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

What is the 7th multiple of 8?

Answer: 56

8, 16, 24, 32, 40, 48, 56; or $7 \times 8 = 56$.

Prerequisite skill: Skip-count to list the multiples of a number

TASK 2 • FOUNDATION 2

Compute 9×12 .

Answer: 108

$9 \times 10 = 90$ and $9 \times 2 = 18$, so 108.

Prerequisite skill: Recall multiplication facts through 12×12

TASK 3 • FOUNDATION 3

Does 6 divide 45 evenly? Explain how you know.

Answer: No. $6 \times 7 = 42$ and $6 \times 8 = 48$, so $45 \div 6$ leaves a remainder of 3. 45 is not a multiple of 6.

A number divides evenly only when the quotient is a whole number with no remainder.

Prerequisite skill: Decide whether one number divides another evenly

TASK 4 • FOUNDATION 4

Write 24 as a product of prime factors.

Answer: $24 = 2 \times 2 \times 2 \times 3$

$24 = 2 \times 12 = 2 \times 2 \times 6 = 2 \times 2 \times 2 \times 3$.

Prerequisite skill: Write a whole number as a product of prime factors

Lesson 6-12: Least Common Multiple

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

List the first 5 multiples of 6. 8×7 ? Does 4 divide 30 evenly?

Answer: 6, 12, 18, 24, 30. $8 \times 7 = 56$. No: $30 \div 4 = 7$ with remainder 2.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

List the first 5 multiples of 8. 9×6 ? Does 3 divide 42 evenly?

Answer: 8, 16, 24, 32, 40. $9 \times 6 = 54$. Yes: $42 \div 3 = 14$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

List the first 5 multiples of 12. 7×12 ? Does 6 divide 40 evenly?

Answer: 12, 24, 36, 48, 60. $7 \times 12 = 84$. No: $40 \div 6 = 6$ with remainder 4.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Cups come in stacks of 6 and lids in packs of 8. What is the smallest number of cups and lids that match exactly, and how many stacks and packs is that?

Answer: LCM of 6 and 8 is 24. That is $24 \div 6 = 4$ stacks of cups and $24 \div 8 = 3$ packs of lids.

Check the result against the situation and explain your reasoning.

Lesson 6-13: Prime Factorization

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute 7×9 .

Answer: 63

$7 \times 9 = 63$, so 63 splits into the factor pair 7 and 9.

Prerequisite skill: Recall multiplication facts through 12×12 to split a number into a factor pair

TASK 2 • FOUNDATION 2

Is 51 prime or composite? Give a factor pair as evidence.

Answer: Composite: $51 = 3 \times 17$.

The digit sum $5 + 1 = 6$ is divisible by 3, so 3 divides 51 and $51 = 3 \times 17$.

Prerequisite skill: Decide whether a number within 100 is prime or composite

TASK 3 • FOUNDATION 3

135 ends in 5, so 5 is a factor. Compute $135 \div 5$.

Answer: 27

$135 \div 5 = 27$, so $135 = 5 \times 27$.

Prerequisite skill: Test divisibility by 2, 3 and 5 and divide out that factor

TASK 4 • FOUNDATION 4

Evaluate $2^3 \times 3$.

Answer: 24

$2^3 = 2 \times 2 \times 2 = 8$, and $8 \times 3 = 24$.

Prerequisite skill: Evaluate a power such as 2^3 as repeated multiplication

Lesson 6-13: Prime Factorization

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Is 39 prime or composite? 6×7 ? Evaluate 2^3 .

Answer: $39 = 3 \times 13$, so 39 is composite. $6 \times 7 = 42$. $2^3 = 2 \times 2 \times 2 = 8$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Is 57 prime or composite? 8×9 ? Evaluate 3^2 .

Answer: $57 = 3 \times 19$, so it is composite. $8 \times 9 = 72$. $3^2 = 3 \times 3 = 9$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Is 43 prime or composite? 7×12 ? Evaluate 5^2 .

Answer: 43 has only the factors 1 and 43, so it is prime. $7 \times 12 = 84$. $5^2 = 5 \times 5 = 25$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Find the smallest whole number whose prime factorization uses 2, 3 and 5, each exactly twice.

Answer: $2^2 \times 3^2 \times 5^2 = 4 \times 9 \times 25 = 900$.

Check the result against the situation and explain your reasoning.

Lesson 6-14: The Distributive Property

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $8 \times 40 + 8 \times 3$.

Answer: 344

$8 \times 40 = 320$ and $8 \times 3 = 24$; $320 + 24 = 344$.

Prerequisite skill: Multiply a one-digit number by a multiple of ten and add the partial products

TASK 2 • FOUNDATION 2

What does $4n$ mean? Write it as a product and describe it in words.

Answer: $4 \times n$: four groups of n .

A number next to a variable means multiplication.

Prerequisite skill: Read a term such as $4n$ as 4 times n

TASK 3 • FOUNDATION 3

Find the greatest common factor of 24 and 36.

Answer: 12

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. The greatest shared factor is 12.

Prerequisite skill: Find the greatest common factor of two whole numbers

TASK 4 • FOUNDATION 4

A rectangle is 6 units tall. Its width is split into parts of 10 and 4 units. Add the two part areas to find the total area.

Answer: 84

$6 \times 10 = 60$ and $6 \times 4 = 24$; $60 + 24 = 84$ square units.

Prerequisite skill: Split a rectangle into two parts and add their areas

Lesson 6-14: The Distributive Property

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$9 \times 30 + 9 \times 4$? GCF of 12 and 18? What does $5y$ mean?

Answer: $9 \times 30 + 9 \times 4 = 270 + 36 = 306$. GCF of 12 and 18 = 6. $5y$ means $5 \times y$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$7 \times 50 + 7 \times 3$? GCF of 16 and 24? What does $8m$ mean?

Answer: $350 + 21 = 371$. The GCF is 8. $8m$ means $8 \times m$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$6 \times 70 + 6 \times 2$? GCF of 15 and 25? What does $2k$ mean?

Answer: $420 + 12 = 432$. The GCF is 5. $2k$ means $2 \times k$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Write $42 + 56$ as a product using the greatest common factor and the distributive property.

Answer: The GCF is 14: $42 + 56 = 14(3 + 4) = 14 \times 7 = 98$.

Check the result against the situation and explain your reasoning.

Lesson 6-15: Simplify Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Name the coefficient and the constant in $9k + 4$.

Answer: The coefficient is 9 and the constant is 4.

9 multiplies k ; 4 has no variable.

Prerequisite skill: Name the coefficient, variable and constant in an expression

TASK 2 • FOUNDATION 2

In the term p , what is the coefficient?

Answer: 1

p means $1 \times p$, so the coefficient is 1.

Prerequisite skill: Read a variable written alone, such as a , as $1a$

TASK 3 • FOUNDATION 3

Expand $3(2w + 5)$.

Answer: $6w + 15$

$3 \times 2w = 6w$ and $3 \times 5 = 15$.

Prerequisite skill: Expand an expression with the distributive property

TASK 4 • FOUNDATION 4

Evaluate $4x + 7$ when $x = 3$.

Answer: 19

$4 \times 3 = 12$ and $12 + 7 = 19$.

Prerequisite skill: Substitute a value into an expression to check equivalence

Lesson 6-15: Simplify Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Coefficient of y in $y + 8$? Expand $2(3a + 4)$.
Evaluate $5x + 1$ when $x = 4$.

Answer: The coefficient is 1, because y means 1 y . $2(3a + 4) = 6a + 8$. $5 \times 4 + 1 = 21$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Coefficient of m in $m - 3$? Expand $4(2b + 1)$.
Evaluate $3x + 2$ when $x = 5$.

Answer: 1. $4(2b + 1) = 8b + 4$. $3 \times 5 + 2 = 17$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Coefficient of k in $7 + k$? Expand $5(c + 3)$.
Evaluate $6x + 4$ when $x = 2$.

Answer: 1. $5(c + 3) = 5c + 15$. $6 \times 2 + 4 = 16$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Simplify $3(2p + 4) + 2(p + 1)$, then evaluate when $p = 3$.

Answer: $6p + 12 + 2p + 2 = 8p + 14$. When $p = 3$: $24 + 14 = 38$.

Check the result against the situation and explain your reasoning.